Data-Driven Menu Optimization and Revenue Concentration Analysis in Specialty Coffee Retail: A Case Study of Afficionado Coffee Roasters

Student Intern
Department of Data Science and Business Analytics
University / Institution Name
City, Country
Email: intern@institution.edu

***Abstract*—Specialty coffee retail businesses operate in highly competitive environments with thin margins and complex product portfolios. Decisions regarding menu composition, inventory allocation, and product discontinuation are historically driven by intuition rather than rigorous quantitative metrics. This paper presents a comprehensive product intelligence framework utilizing transaction-level retail analytics from Afficionado Coffee Roasters, covering 149,116 transactions across three urban storefronts in 2025. Through empirical revenue computation, volume-to-value rank analysis, Pareto evaluation, and an algorithmic SKU Efficiency Score (SES), we expose severe operational inefficiencies. Our results reveal a total realized revenue of \$698,812.33 from 214,470 units sold across 80 unique SKUs. A modified Pareto concentration check proves that 53.75% of the product menu anchors 80% of aggregate revenue. Furthermore, categories such as Loose Tea and Packaged Chocolate exhibit critical inefficiency, while Drinking Chocolate showcases maximum revenue velocity per SKU (\$18,104.00). This paper translates these empirical data insights into structural menu-engineering frameworks, laying the foundation for automated Streamlit dashboards and objective product pruning strategies.**

***Index Terms*—Retail Analytics, Menu Optimization, Pareto Analysis, SKU Efficiency, Product Intelligence, Data-Driven Decision Making.**

I. Introduction

Modern retail operations generate vast quantities of transactional and product-level data. However, translating raw data into structural menu optimization strategies remains a substantial barrier for small-to-medium enterprises (SMEs) like Afficionado Coffee Roasters. Without granular analysis, menu architectures expand arbitrarily, introducing hidden costs via supply chain complexity, specialized storage demands, and cognitive fatigue for the consumer.

Afficionado Coffee Roasters possesses high-fidelity transactional logs but faces three acute structural visibility gaps:

- **Asymmetric Visibility:** Ambiguity regarding whether a product's popularity (sales volume) correlates cleanly with its financial profitability (revenue contribution).
- **Category Dependency Risk:** A lack of quantification regarding how heavily overall business viability rests on core versus peripheral product segments.
- **Long-Tail Overhead:** Inability to isolate low-impact, underperforming menu entries that occupy retail space without driving proportional growth.

To address these vulnerabilities, this study establishes an empirical product intelligence pipeline. The primary objectives are to mathematically rank sales performance, map category dependency, and model item-level efficiency. By shifting the paradigm from *when* customers purchase to *what* drives systemic revenue velocity, this paper outlines actionable pathways for catalog rationalization and strategic category design.

II. Data and Analytical Methodology

The analytical framework adheres to a structured, reproducible five-stage lifecycle: Data Ingestion and Validation, Revenue Computation, Product Popularity Metrics, Category Dependency Analysis, and Menu Concentration Profiling.

A. Data Source and Schema Validation

The raw dataset consists of $N = 149,116$ row records documenting real-time coffeehouse sales for the calendar year 2025. The data schema comprises 11 descriptive fields across categorical and numeric domains, including unique identifiers (transaction_id, product_id), spatio-temporal logs (transaction_time, store_location), and product characteristics (product_category, product_type, product_detail). Preliminary data validation confirmed structural integrity: zero missing values in essential fields, positive non-zero constraints for unit price ($P_i > 0$), and positive quantity integers ($Q_j \geq 1$).

B. Mathematical Formulations and Revenue Computation

For any distinct transaction index $j \in \{1, 2, \ldots, N\}$, individual transactional revenue is determined as the inner dot product of unit price and purchase quantity:

$$\text{Revenue}_j = \text{transaction\_qty}_j \times \text{unit\_price}_j \tag{1}$$

Let $\mathcal{P}$ denote the set of all unique stock-keeping units (SKUs), where $|\mathcal{P}| = 80$. For a given product $i \in \mathcal{P}$, total revenue (R_i) and total sales volume (V_i) are calculated by aggregating over the subset of transactions $\mathcal{T}_i$ where product i was purchased:

$$R_i = \sum_{j \in \mathcal{T}_i} (\text{transaction\_qty}_j \times \text{unit\_price}_j) \tag{2}$$

$$V_i = \sum_{j \in \mathcal{T}_i} \text{transaction\_qty}_j \tag{3}$$

The aggregate systemic revenue (R_{total}) is defined as:

$$R_{\text{total}} = \sum_{i \in \mathcal{P}} R_i \tag{4}$$

The relative revenue share (S_i) of a unique item expressed as a percentage is:

$$S_i = \left(\frac{R_i}{R_{\text{total}}} \right) \times 100 \tag{5}$$

C. Product Efficiency Matrix

To prevent macro-level category totals from distorting internal item performance, we introduce the *SKU Efficiency Score* (SES_c) for any product category $c \in \mathcal{C}$. Let $\mathcal{P}_c \subseteq \mathcal{P}$ represent the set of SKUs assigned to category c. The score normalizes category revenue against its catalog breadth:

$$SES_c = \frac{\sum_{i \in \mathcal{P}_c} R_i}{|\mathcal{P}_c|} \tag{6}$$

A low SES_c coupled with high cardinality $|\mathcal{P}_c|$ denotes a highly fragmented category that demands intensive catalog pruning.

D. Rank Discrepancy Modeling

To evaluate the decoupling of item popularity from revenue importance, each product is assigned a volume rank ($\text{Rank}_{V,i}$) and a revenue rank ($\text{Rank}_{R,i}$), sorted in descending order from 1 to $|\mathcal{P}|$. The Rank Discrepancy Index (Δ_i) is mathematically structured as:

$$\Delta_i = \text{Rank}_{R,i} - \text{Rank}_{V,i} \tag{7}$$

A highly negative Δ_i highlights high-margin, low-volume asset plays, whereas a large positive Δ_i uncovers low-margin volume anchors.

III. Experimental Results and Analysis

Execution of the data pipeline generated a total realized business revenue of **$698,812.33** and an aggregate sales volume of **214,470 items** across 2025.

A. Geographic and Store-Level Distribution

An inspection of physical store footprints indicates an exceptionally balanced revenue distribution across regions. The sales metrics per store are detailed in Table I. The maximum absolute revenue variance between the highest-performing location (Hell's Kitchen) and the lowest (Lower Manhattan) is merely 0.92%, demonstrating uniform brand adoption across geographic sub-markets.

TABLE I
PERFORMANCE SUMMARY BY STORE LOCATION

Store Location	**Units Sold**	**Revenue ($)**	**Share (%)**
Hell's Kitchen	71,737	236,511.17	33.84%
Astoria	70,991	232,243.91	33.23%
Lower Manhattan	71,742	230,057.25	32.92%

B. Category Performance Metrics and Efficiency Mapping

Aggregating the transactional rows into the nine major product super-groups exposes substantial asymmetry in structural dependence. Table II breaks down the volume, revenue, SKU counts, and structural SKU Efficiency Scores.

The empirical data indicates that **Coffee** and **Tea** together serve as the organizational baseline, driving 66.74% of all gross revenues and accounting for 74.13% of volume demand. However, the SES metric isolates **Drinking Chocolate** as the most hyper-efficient category on the menu: with only 4 registered SKUs, it brings in $72,416.00, yielding a peak efficiency of $18,104.00 per SKU. Conversely, **Loose Tea** and **Packaged Chocolate** display critical weakness, generating an SES of only $1,401.70 and $1,469.21 respectively, highlighting a bloated catalog framework that yields minimal financial velocity.

C. Product-Level Performance and Rank Discrepancies

Drilling down to individual item performance highlights the explicit leaders and trailing products within the menu ecosystem. Table III arrays the top 5 and bottom 5 items ranked purely by total dollar revenue generation.

Analyzing the Rank Discrepancy Index (Δ_i) reveals critical architectural insights for menu design. The most profound positive and negative rank deltas are captured explicitly:

- **High-Margin Premium Plays ($\Delta_i \ll 0$):** *Civet Cat Coffee Beans* holds a revenue rank of 27 but a volume rank of 58 ($\Delta = -31$), powered by a premium unit price of $45.00. Similarly, *Cappuccino Lg* scores a revenue rank of 4 while holding a volume rank of 35 ($\Delta = -31$, Price = $4.25). These products represent powerful margin enhancers that drive financial strength despite lower transaction volumes.
- **Low-Margin Volume Anchors ($\Delta_i \gg 0$):** *Earl Grey Rg* holds the absolute absolute position for unit volume sold ($\text{Rank}_V = 1$), yet registers a secondary revenue position at $\text{Rank}_R = 26$ ($\Delta = +25$, Price = $2.50). This trend is shared by *Our Old Time Diner Blend Sm* ($\Delta = +32$) and *Morning Sunrise Chai Rg* ($\Delta = +26$). These items represent loss leaders or high-frequency baseline traffic drivers.

D. Revenue Concentration and Pareto Verification

To measure the operational risk and dependency structure of the establishment, a cumulative pareto distribution check was run across the sorted catalog of 80 items. The analysis indicates that **43 out of 80 products (53.75%)** are required to hit the 80% threshold of cumulative revenue ($559,049.86).

TABLE II
COMPREHENSIVE CATEGORY-LEVEL BREAKDOWN AND SKU EFFICIENCY SCORE (SES)

Product Category	**Units Sold**	**Unit Share (%)**	**Revenue ($)**	**Revenue Share (%)**	**SKU Count**	**SES ($ / SKU)**
Coffee	89,250	41.61%	269,952.45	38.63%	21	12,854.88
Tea	69,737	32.52%	196,405.95	28.11%	16	12,275.37
Bakery	23,214	10.82%	82,315.64	11.78%	11	7,483.24
Drinking Chocolate	17,457	8.14%	72,416.00	10.36%	4	18,104.00
Coffee Beans	1,828	0.85%	40,085.25	5.74%	10	4,008.53
Branded Merchandise	776	0.36%	13,607.00	1.95%	3	4,535.67
Loose Tea	1,210	0.56%	11,213.60	1.60%	8	1,401.70
Flavours	10,511	4.90%	8,408.80	1.20%	4	2,102.20
Packaged Chocolate	487	0.23%	4,407.64	0.63%	3	1,469.21

TABLE III
GRANULAR ITEM PERFORMANCE: TOP 5 VS. BOTTOM 5 SKUS BY TOTAL REVENUE

Product Detail Name	**Product Category**	**Units Sold**	**Revenue ($)**	**Rev Share (%)**	**Rev Rank**	**Vol Rank**
Sustainably Grown Organic Lg	Drinking Chocolate	4,453	21,151.75	3.03%	1	11
Dark chocolate Lg	Drinking Chocolate	4,668	21,006.00	3.01%	2	2
Latte Rg	Coffee	4,497	19,112.25	2.73%	3	8
Cappuccino Lg	Coffee	4,151	17,641.75	2.52%	4	35
Morning Sunrise Chai Lg	Tea	4,346	17,384.00	2.49%	5	19
Lemon Grass	Loose Tea	152	1,360.40	0.19%	76	72
Guatemalan Sustainably Grown	Coffee Beans	134	1,340.00	0.19%	77	78
Spicy Eye Opener Chai	Loose Tea	122	1,335.90	0.19%	78	79
Earl Grey	Loose Tea	142	1,270.90	0.18%	79	77
Dark chocolate	Packaged Chocolate	118	755.20	0.11%	80	80

While a classic, highly skewed Pareto structure (e.g., 20% of items driving 80% of revenue) is not present here, the results demonstrate a strong, extended "long-tail" signature. The bottom 20 items combined contribute less than 4% of aggregate revenues, indicating significant menu bloat in peripheral variants that add operational complexity without moving financial metrics.

IV. DISCUSSION AND STRATEGIC MENU RECOMMENDATIONS

The integration of these analytical findings allows us to move away from intuition-driven management and implement a structured **Menu Engineering Framework**:

A. Catalog Rationalization and Pruning

The primary candidates for structural elimination are localized within the **Loose Tea** and **Packaged Chocolate** categories. Loose Tea features a low efficiency score ($SES = \$1,401.70$) spread across 8 separate SKUs, with individual products like *Earl Grey (Loose)*, *Spicy Eye Opener Chai (Loose)*, and *Lemon Grass (Loose)* each failing to cross even 0.2% of revenue share. Eliminating loose-leaf retail options simplifies back-of-house storage, eliminates stagnant inventory write-offs, and shifts traffic toward high-velocity liquid alternatives.

B. Capitalizing on High-Efficiency "Hero" Products

Drinking Chocolate is identified as a critical revenue driver. Specifically, *Sustainably Grown Organic Lg* and *Dark chocolate Lg* stand as the top two grossing products in the entire establishment ($21,151.75 and $21,006.00 respectively). Management should expand marketing visibility for these items, introduce seasonal variants, and establish premium bundles (e.g., pairing hot chocolates with high-margin Bakery items).

C. Price Optimization and Volume Transitions

The extreme volume traction of items like *Earl Grey Rg* (Vol Rank $=$ 1, Rev Rank $=$ 26) confirms that customers possess immense baseline demand for core brewed teas. Because these items have highly inelastic demand characteristics as morning staples, executing a modest, data-justified price adjustment of $0.15 to $0.25 on regular-sized brewed teas would directly improve gross margins without damaging volume velocity.

V. CONCLUSION AND FUTURE WORK

This study replaces intuition-based menu design at Afficionado Coffee Roasters with a rigorous data science architecture. By processing 149,116 transaction histories, we successfully isolated top-tier premium anchors, exposed a bloated long-tail catalog in specialized sub-categories, and calculated precise SKU efficiency distributions.

As a direct progression of this internship initiative, these structural analytical parameters have been formalized into a multi-module **Streamlit Dashboard web application**. The software tool dynamically presents the volume-to-revenue scatter fields, dynamically slices distributions using Top-N product sliders, and offers multi-store filtering across physical hubs. Future iterations will integrate predictive time-series modeling (e.g., Prophet or ARIMA formulations) to project ingredient depletion rates, enabling automated inventory procurement aligned directly with empirical demand cycles.

References

[1] M. Kasavana and D. Smith, *Menu Engineering: A Practical Guide to Rice and Menu Analysis*. Corvallis, OR: Edgell Communications, 1990.

[2] P. Jones and M. Comfort, “Data-driven decision making in the hospitality industry,” *International Journal of Hospitality Management*, vol. 39, pp. 122–130, 2014.

[3] V. Paret, *The 80/20 Principle: The Secret to Achieving More with Less*. New York: Currency, 1999.

VI. Executive Summary for Government Stakeholders

This section serves as a high-level briefing for government, regulatory, and public institutional stakeholders to evaluate the economic footprint, resilience, and industrial compliance framework of Afficionado Coffee Roasters’ regional market operations.

A. Macro-Economic Impact and Market Footprint

During the fiscal year 2025, regional operations generated a total realized direct economic output of **$698,812.33**, driven by the distribution of **214,470 consumer units**. Analysis across physical retail environments showcases an exceptionally uniform economic distribution:

- **Hell’s Kitchen:** $236,511.17 (33.84% gross output)
- **Astoria:** $232,243.91 (33.23% gross output)
- **Lower Manhattan:** $230,057.25 (32.92% gross output)

The baseline variance across municipalities is negligible ($< 1\%$), highlighting regional economic stability, balanced consumer retail spending, and dependable local commercial contribution across urban districts.

B. Risk Diversification and Systemic Resilience

A core mandate for economic regulatory bodies is checking business vulnerability to supply shocks or market changes. A modified Pareto distribution metric was deployed across the enterprise’s catalog of 80 unique stock-keeping units (SKUs).

The empirical findings confirm that **53.75% of the product portfolio** (43 active SKUs) is required to sustain 80% of the aggregate business revenue. This structure represents low exposure to portfolio concentration risks compared to standard retail archetypes that suffer from single-product dependency loops. The operational structure indicates that the enterprise is highly resilient against localized agricultural or commodity supply chain disruptions.

C. Strategic Interventions for Regional Optimization

To optimize resource usage and improve commercial sustainability, public stakeholders should support the following data-justified corporate changes:

1) **Supply Chain Rationalization:** Phasing out low-velocity peripheral segments such as *Loose Tea* (which presents a weak efficiency score of only $1,401.70 per SKU across 8 items) to reduce municipal waste and eliminate stagnant warehouse overhead.
2) **Infrastructure for High-Efficiency Nodes:** Incentivizing production lines around high-velocity segments like *Drinking Chocolate*, which exhibits an optimal efficiency profile of $18,104.00 per active SKU.

D. Integration of Open Traceability Frameworks

As an alignment milestone with contemporary public policy initiatives on food transparency and open data, the enterprise’s analytics structure is migrating toward an integrated system architecture. By deploying live analytical frameworks alongside decentralized logging protocols, the business architecture enables transparent tracking and verifiable visibility parameters from supply networks directly down to individual point-of-sale customer receipts. This capability directly reduces friction for corporate audits, ensures food safety compliance, and establishes a template for transparency within the regional agri-food and retail tech ecosystems.